0/1 knapsack problem using dynamic programming

Given

item	wt	Value
1	2	₹12
2	1	₹10
3	3	₹20
4	2	₹15

Capacity (W) = 5

Capacity → j

items ↓ i	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	12	12	12	12
2	0	10	12	22	22	22
3	0	10	12	22	30	32
4	0	10	15	25	30	37

To find F(1,1) $i=1, j=1$

if $j < wt_i$

$1 < 2$

Value ← $F(0,1) = 0$

$F(1,1) = 0$

To find F(1,2)

if $2 \nless 2$

$$\text{Value} \leftarrow \max \begin{cases} F(0,2) = 0 \\ 12 + F(0,0) = 12 \end{cases}$$

$F(1,2) = 12$

To find F(1,3)

if $j < wt_i$

$3 \nless 2$

$$\text{Value} \leftarrow \max \begin{cases} F(0,3) = 0 \\ 12 + F(0,1) = 12 \end{cases}$$

To find F(1,4)

if $j < wt_i$

$4 \nless 2$

$$\text{Value} = \max \begin{cases} F(0,4) = 0 \\ 12 + F(0,2) = 12 \end{cases}$$

$F(1,4) = 12$

To find F(1,5)

if $j < wt_i$

$5 \nless 2$

$$\text{Value} \leftarrow \max \begin{cases} F(0,5) = 0 \\ 12 + F(0,3) = 12 \end{cases}$$

$F(1,5) = 12$

To find $F(2,1)$

if $j < wt_i$

$1 \not< 1$

$$\text{Value} \leftarrow \max \begin{cases} F(1,1) = 0 \\ 10 + F(1,0) = 10 \end{cases}$$

$F(2,1) = 10$

To find $F(2,2)$

if $j < wt_i$

$2 \not< 1$

$$\text{Value} \leftarrow \max \begin{cases} F(1,2) = 12 \checkmark \\ 10 + F(1,1) = 10 \end{cases}$$

$F(2,2) = 12$

To find $F(2,3)$

if $j < wt_i$

$3 \not< 1$

$$\text{Value} \leftarrow \max \begin{cases} F(1,3) = 12 \\ 10 + F(1,2) = 22 \checkmark \end{cases}$$

To find $F(2,4)$

if $j < wt_i$

$4 \not< 1$

$$\text{Value} = \max \begin{cases} F(1,4) = 12 \\ 10 + F(1,3) = 10 + 12 = 22 \checkmark \end{cases}$$

$F(2,4) = 22$

To find $F(2,5)$

if $j < wt_i$

$5 \not< 1$

$$\text{Value} \leftarrow \max \begin{cases} F(1,5) = 12 \\ \boxed{10 + F(1,4) = 10 + 12 = 22 \checkmark} \end{cases}$$

To find $F(3,1)$

if $j < wt_i$

$1 < 3$

$F(3,1) = F(2,1) = 10$

To find $F(3,2)$

if $j < wt_i$

$2 < 3$

$F(3,2) = F(2,2) = 12$

To find $F(3,3)$

if $j < wt_i$

$3 \not< 3$

$$F(3,3) = \max \begin{cases} F(2,3) = 22 \checkmark \\ 20 + F(1,0) = 20 \end{cases}$$

To find $F(3,4)$

if $4 \not< wt_i\ (3)$

$$F(3,4) = \max \begin{cases} F(2,4) = 22 \\ 20 + F(2,1) = 30 \checkmark \end{cases}$$

To find $F(3,5)$

if $5 \not< 3$

$$F(3,5) = \max \begin{cases} F(2,5) = 22 \\ 20 + F(2,2) = 32 \checkmark \end{cases}$$

To find $F(4,1)$

if $j < wt_i$

$1 < 2$

$F(4,1) = F(3,1) = 10$

To find $F(4,2)$

if $j < wt_i$

$2 \not< 2$

$$F(4,2) = \max \begin{cases} F(3,2) = 12 \\ 15 + F(3,0) = 15 \checkmark \end{cases}$$

To find $F(4,3)$

if $j < wt_i$

$3 \not< 2$

$$F(4,3) = \max \begin{cases} F(3,3) = 22 \\ 15 + F(3,1) = 25 \checkmark \end{cases}$$

To find $F(4,4)$

if $j < wt_i$

$4 \not< 2$

$$F(4,4) = \max \begin{cases} F(3,4) = 30 \checkmark \\ 15 + F(3,2) = 27 \end{cases}$$

To find $F(4,5)$

if $j < wt_i$

$5 \not< 2$

$$F(4,5) = \max \begin{cases} F(3,5) = 32 \\ 15 + F(3,3) = 37 \checkmark \end{cases}$$

To find which items are to be included in knapsack bag

$F(4,5) = 37$ (Optimal profit value)

To decide whether to add 4^{th} item

$F(4,5) \neq F(3,5)$

$37 \neq 32$

So include 4^{th} item in knapsack bag.

Now $F(4,5)$ becomes $F(i-1, j-w_i)$. So $F(3,5-2) =$

$F(3,3) = F(2,3)$

$22 = 22$

So don't include 3^{rd} item. So $F(3,3)$ becomes $F(i-1, j) = F(2,3)$

$F(2,3) \neq F(1,3)$

$22 \neq 12$

So include 2^{nd} item in knapsack bag. So $F(2,3)$ becomes $F(i-1, j-w_i) \Rightarrow F(1,2)$

$F(1,2) = F(0,2)$

$12 \neq 0$

So include 1^{st} item in knapsack bag.

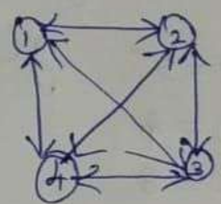
Soln

Include $1 \rightarrow 2 \rightarrow 4$ in knapsack bag to get Optimal profit value $= 37$

Problem

The cost adjacency matrix for the foll. graph which marks various locations a traveling salesman visits, as vertices is given below.

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 5 & 4 & 3 & 2 \\ 2 & 2 & 3 & 3 & 1 \\ 3 & 2 & 6 & 1 & 3 \\ 4 & 3 & 6 & 5 & 2 \end{array}$$



The recursive definition for <u>traveling salesman problem</u> is given below.

$$g(i,S) = \min_{k \in S} \{ c_{ik} + g(k, S - \{k\}) \}$$

where 'i' is the starting location of the traveling salesman and 'S' gives the remaining vertices excluding starting location 'i'. We try to find the shortest possible route that passes through all locations once and return to the starting location.

$$g(1, \{2,3,4\}) = \min_{k \in \{2,3,4\}} \{ c_{1k} + g(k, \{2,3,4\} - \{k\}) \}$$

$g(1,\{2,3,4\})$ = 12

i = 1

k = 2: $c_{12} + g(2, \{3,4\})$ = 4 + 8 = 12

- i = 2, k = 3: $c_{23} + g(3, \{4\})$ = 3 + 6 = 9
  - k = 4: $c_{34} + g(4, \phi)$ = 3 + c_{41} = 6
- i = 2, k = 4: $c_{24} + g(4, \{3\})$ = 1 + 7 = 8
  - k = 3: $c_{43} + g(3, \phi)$ = 5 + c_{31} = 7

k = 3: $c_{13} + g(3, \{2,4\})$ = 3 + 10 = 13

- i = 3, k = 2: $c_{32} + g(2, \{4\})$ = 6 + 4 = 10
  - k = 4: $c_{24} + g(4, \phi)$ = 1 + c_{41} = 4
- i = 3, k = 4: $c_{34} + g(4, \{2\})$ = 3 + 8 = 11
  - k = 2: $c_{42} + g(2, \phi)$ = 6 + c_{21} = 8

k = 4: $c_{14} + g(4, \{2,3\})$ = 2 + 11 = 13

- k = 2: $c_{42} + g(2, \{3\})$ = 6 + 5 = 11
  - k = 3: $c_{23} + g(3, \phi)$ = 3 + c_{31} = 5
- k = 3: $c_{43} + g(3, \{2\})$ = 5 + 8 = 13
  - k = 2: $c_{32} + g(2, \phi)$ = 6 + c_{21} = 8

<u>Soln</u>

1 → 2 → 4 → 3 → 1 is the shortest route traveling salesman should take to get minimum distance covered equal to 12

$g(2, \phi) = 2$ $\quad g(2, \{3, 4\}) = 8$

$g(3, \phi) = 2$ $\quad g(3, \{2, 4\}) = 10$

$g(4, \phi) = 3$ $\quad g(4, \{2, 3\}) = 11$

$g(2, \{3\}) = 5$ $\quad g(1, \{2, 3, 4\}) = 12$

$g(2, \{4\}) = 4$

$g(3, \{2\}) = 8$

$g(3, \{4\}) = 6$

$g(4, \{2\}) = 8$

$g(4, \{3\}) = 7$